Combined Team Weekly Reports (all 5 members)

Smart Hospital Management System (SmartCare) | Team Nobita (PBL2627-AI&DS-A-031;) | Mentor: Er. Ram Babu Buri

One report per week for the whole team, 13 weeks from 06-07-2026 to 04-10-2026. Paste the three boxes into the Weekly Report tab: Work Completed This Week, Problems Faced and Plan for Next Week. GitHub link for all weeks: https://github.com/mathurdevansh/Smart-Hospital-Management-System

Member	Module
Devansh Mathur	Module 1 - Authentication, Admin and Security
Deepak Kumar	Module 2 - Patient and Doctor Management
Hansraj Pandey	Module 3 - Appointment and Consultation
Aryan Daware	Module 4 - Pharmacy, Laboratory and Nursing
Devkinandan Kumawat	Module 5 - Billing, Admission, Reports and Feedback

Week 1 | 06-07-2026 to 12-07-2026 | Team formation, guide selection and abstract

Week dropdown	Week 1
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	This week the whole team the team started the project and finalised its idea. Devansh (Authentication): joined team of five, discussed project ideas and shortlisted a hospital management system; studied problems in manual hospital record keeping and listed login and role requirements; prepared the abstract section on authentication, roles and admin control. Deepak (Patient): joined team of five and discussed hospital problems the project could solve; collected how patients and doctors are registered in small hospitals; wrote the abstract section on patient and doctor management. Hansraj (Appointment): joined team of five and listed hospital workflows to digitise; studied how appointment booking is handled in clinics; wrote the abstract section on appointments and consultation. Aryan (Pharmacy): joined team of five and brainstormed hospital system ideas; studied pharmacy and laboratory handling in hospitals; wrote the abstract section on prescriptions, lab tests and vitals. Devkinandan (Billing): joined team of five and discussed the project idea together; studied how hospital billing and admission work; wrote the abstract section on billing, admission and reports. All members shared their work on GitHub and discussed it in the team and in the Saturday mentor meeting.
Problems Faced	Devansh: Difficulty in choosing one idea among several. Deepak: Team had many different ideas. Hansraj: Choosing the right scope for the idea. Aryan: Too many ideas to compare. Devkinandan: Agreeing on a single project scope. All problems were solved by team discussion and guidance from the mentor.
Plan for Next Week	Next week the team will move to: SRS (Software Requirement Specification). Devansh: identified five user roles: Admin, Doctor, Patient, Nurse and Receptionist. Deepak: listed patient requirements: registration, profile, history and records. Hansraj: listed functional requirements for booking, rescheduling and cancelling. Aryan: listed requirements for medicine stock and expiry tracking. Devkinandan: listed requirements for bill generation and payments.

Week 2 | 13-07-2026 to 19-07-2026 | SRS (Software Requirement Specification)

Week dropdown	Week 2
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	This week the whole team the focus was on the Software Requirement Specification. Devansh (Authentication): identified five user roles: Admin, Doctor, Patient, Nurse and Receptionist; wrote functional requirements for login, logout, session handling and role-based access; wrote security requirements: password hashing, session timeout, audit trail. Deepak (Patient): listed patient requirements: registration, profile, history and records; listed doctor requirements: profile, department, specialisation and availability; wrote data requirements for patient and doctor details. Hansraj (Appointment): listed functional requirements for booking, rescheduling and cancelling; wrote requirements for doctor consultation and diagnosis notes; defined appointment statuses: Scheduled, Completed and Cancelled. Aryan (Pharmacy): listed requirements for medicine stock and expiry tracking; listed requirements for doctor prescriptions with medicine items; listed requirements for lab test request and result entry. Devkinandan (Billing): listed requirements for bill generation and payments; listed requirements for room allocation and admission; listed requirements for admin reports and revenue summary. All members shared their work on GitHub and discussed it in the team and in the Saturday mentor meeting.
Problems Faced	Devansh: Unclear which roles needed separate dashboards. Deepak: Overlap between patient and receptionist features. Hansraj: Defining all appointment statuses clearly. Aryan: Many pharmacy rules to capture. Devkinandan: Unclear billing rules at the start. All problems were solved by team discussion and guidance from the mentor.
Plan for Next Week	Next week the team will move to: UML design (use-case, class, sequence). Devansh: drew the use-case diagram for the Admin actor (manage users, departments, audit logs). Deepak: drew the use-case diagram for Patient and Doctor actors. Hansraj: drew the use-case diagram for booking and consultation. Aryan: drew the use-case diagram for Nurse and Lab actors. Devkinandan: drew the use-case diagram for billing and admission.

Week 3 | 20-07-2026 to 26-07-2026 | UML design (use-case, class, sequence)

Week dropdown	Week 3
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	This week the whole team the focus was on UML design diagrams. Devansh (Authentication): drew the use-case diagram for the Admin actor (manage users, departments, audit logs); drew the login and logout sequence diagram with filters and session; designed class diagram for User, Role and AuditLog. Deepak (Patient): drew the use-case diagram for Patient and Doctor actors; designed the class diagram for Patient with attributes and methods; designed the class diagram for Doctor and Receptionist. Hansraj (Appointment): drew the use-case diagram for booking and consultation; designed the class diagram for Appointment; designed the class diagram for MedicalRecord. Aryan (Pharmacy): drew the use-case diagram for Nurse and Lab actors; designed the class diagram for Medicine and MedicineInventory; designed the class diagram for Prescription and PrescriptionItem. Devkinandan (Billing): drew the use-case diagram for billing and admission; designed the class diagram for Bill and Payment; designed the class diagram for Admission and Room. All members shared their work on GitHub and discussed it in the team and in the Saturday mentor meeting.
Problems Faced	Devansh: Mixing class diagram relations for User and Role. Deepak: Deciding attributes for Patient class. Hansraj: Showing the double-booking rule in sequence diagram. Aryan: Parent-child relation for prescription items. Devkinandan: Relation between admission and room. All problems were solved by team discussion and guidance from the mentor.
Plan for Next Week	Next week the team will move to: Database (ER) design, UI mock-ups, project setup. Devansh: designed ER tables roles, users and audit_logs with keys and constraints. Deepak: designed the patients table with personal and contact fields. Hansraj: designed the appointments table with doctor, patient and slot columns. Aryan: designed the medicines and medicine_inventory tables. Devkinandan: designed the billing and payments tables.

Week 4 | 27-07-2026 to 02-08-2026 | Database (ER) design, UI mock-ups, project setup

Week dropdown	Week 4
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	This week the whole team the focus was on database design, mock-ups and project setup. Devansh (Authentication): designed ER tables roles, users and audit_logs with keys and constraints; designed the departments table and linked doctors to departments; created Maven project structure with pom.xml and Jakarta Servlet dependencies. Deepak (Patient): designed the patients table with personal and contact fields; designed the doctors table with department link and qualification; designed the medical_records table and its relation to patients. Hansraj (Appointment): designed the appointments table with doctor, patient and slot columns; designed the medical_records table linked to appointments; worked out indexes for date and doctor lookups. Aryan (Pharmacy): designed the medicines and medicine_inventory tables; designed prescriptions and prescription_items tables; designed lab_tests and patient_vitals tables. Devkinandan (Billing): designed the billing and payments tables; designed the rooms and admissions tables; designed the feedback and notifications tables. All members shared their work on GitHub and discussed it in the team and in the Saturday mentor meeting.
Problems Faced	Devansh: Foreign key order created errors in SQL script. Deepak: Normalising patient and record tables. Hansraj: Choosing correct indexes for lookups. Aryan: Foreign key order in schema. Devkinandan: Choosing data types for amounts. All problems were solved by team discussion and guidance from the mentor.
Plan for Next Week	Next week the team will move to: Module coding - models and DAO layer. Devansh: wrote the Role and User model classes with getters and setters. Deepak: wrote Patient model class with all fields. Hansraj: wrote the Appointment model with date and time fields. Aryan: wrote Medicine and MedicineInventory model classes. Devkinandan: wrote Bill and Payment model classes.

Week 5 | 03-08-2026 to 09-08-2026 | Module coding - models and DAO layer

Week dropdown	Week 5
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	This week the whole team coding of the module began with models and DAO classes. Devansh (Authentication): wrote the Role and User model classes with getters and setters; created DBConnection utility reading db.properties; implemented UserDAO methods to fetch user by username and email. Deepak (Patient): wrote Patient model class with all fields; wrote Doctor and Receptionist model classes; implemented PatientDAO insert and select methods. Hansraj (Appointment): wrote the Appointment model with date and time fields; wrote the MedicalRecord model class; implemented AppointmentDAO insert and fetch by patient. Aryan (Pharmacy): wrote Medicine and MedicineInventory model classes; wrote Prescription and PrescriptionItem models; wrote LabTest and PatientVital models. Devkinandan (Billing): wrote Bill and Payment model classes; wrote Admission and Room model classes; wrote Feedback and Notification models. All members shared their work on GitHub and discussed it in the team and in the Saturday mentor meeting.
Problems Faced	Devansh: Driver not found error for MySQL connector. Deepak: Date conversion between java.util and java.sql. Hansraj: SQL Time type mapping was confusing. Aryan: Mapping decimal fields correctly. Devkinandan: Decimal rounding in totals. All problems were solved by team discussion and guidance from the mentor.
Plan for Next Week	Next week the team will move to: Module coding - services and servlets. Devansh: wrote PasswordUtil using the jBCrypt library for hashing and verification. Deepak: implemented PatientServlet actions for view and update profile. Hansraj: created AppointmentService with input validation. Aryan: implemented MedicineServlet for inventory add and update. Devkinandan: built BillingService to calculate totals and taxes.

Week 6 | 10-08-2026 to 16-08-2026 | Module coding - services and servlets

Week dropdown	Week 6
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	This week the whole team the service and servlet layers of the module were built. Devansh (Authentication): wrote PasswordUtil using the jBCrypt library for hashing and verification; built AuthService login flow with credential and role checking; implemented LoginServlet doGet and doPost with redirect by role. Deepak (Patient): implemented PatientServlet actions for view and update profile; implemented DoctorServlet dashboard action with today's data; added the patient search action by name and phone. Hansraj (Appointment): created AppointmentService with input validation; added the double-booking check inside the service; implemented AppointmentServlet for booking requests. Aryan (Pharmacy): implemented MedicineServlet for inventory add and update; implemented PrescriptionServlet to save multiple medicine items; implemented LabServlet for test request and result update. Devkinandan (Billing): built BillingService to calculate totals and taxes; built AdmissionService for admit, transfer and discharge logic; implemented BillingServlet for bill generation and payment. All members shared their work on GitHub and discussed it in the team and in the Saturday mentor meeting.
Problems Faced	Devansh: Role redirect looped back to login page. Deepak: Null pointer when patient had no record. Hansraj: Double-booking check allowed edge cases. Aryan: Saving many items needed a loop with transaction. Devkinandan: Room status not updated on discharge. All problems were solved by team discussion and guidance from the mentor.
Plan for Next Week	Next week the team will move to: Module coding - JSP views and UI. Devansh: created login.jsp with responsive form and error messages. Deepak: created the patient dashboard.jsp with quick links. Hansraj: created the patient book_appointment.jsp with doctor and slot selection. Aryan: created the inventory.jsp with stock table and add form. Devkinandan: created the invoice.jsp with printable layout.

Week 7 | 17-08-2026 to 23-08-2026 | Module coding - JSP views and UI

Week dropdown	Week 7
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	This week the whole team the JSP pages and user interface were developed. Devansh (Authentication): created login.jsp with responsive form and error messages; created admin dashboard.jsp with summary cards; built admin users.jsp with table for add, edit and delete users. Deepak (Patient): created the patient dashboard.jsp with quick links; created the patient profile.jsp with editable form; created the doctor dashboard.jsp with appointment summary. Hansraj (Appointment): created the patient book_appointment.jsp with doctor and slot selection; created the patient appointments.jsp with status badges; created the receptionist book_appointment.jsp. Aryan (Pharmacy): created the inventory.jsp with stock table and add form; created prescriptions.jsp for the doctor and prescription_view.jsp; created lab_management.jsp and lab_test_details.jsp. Devkinandan (Billing): created the invoice.jsp with printable layout; created receptionist billing.jsp and generate_bill.jsp; created common admit_patient.jsp and admission_list.jsp. All members shared their work on GitHub and discussed it in the team and in the Saturday mentor meeting.
Problems Faced	Devansh: Include files broke relative paths in admin pages. Deepak: Form values lost after validation failure. Hansraj: Date picker accepted past dates. Aryan: Dynamic rows in prescription form. Devkinandan: Invoice print layout broke on paper size. All problems were solved by team discussion and guidance from the mentor.
Plan for Next Week	Next week the team will move to: Module coding - validation, exceptions, security. Devansh: wrote ValidationUtil for email, phone and empty field checks. Deepak: added server-side validation for phone, age and blood group. Hansraj: added past-date and empty reason validations. Aryan: added validation for quantity, dose and vitals ranges. Devkinandan: added validation for amount, payment mode and dates.

Week 8 | 24-08-2026 to 30-08-2026 | Module coding - validation, exceptions, security

Week dropdown	Week 8
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	This week the whole team validation, exception handling and security were added. Devansh (Authentication): wrote ValidationUtil for email, phone and empty field checks; created ApplicationException and DatabaseException classes; implemented AuthenticationFilter to block pages for logged-out users. Deepak (Patient): added server-side validation for phone, age and blood group; added JavaScript validation on registration form fields; handled exceptions in DAO methods with clear user messages. Hansraj (Appointment): added past-date and empty reason validations; handled booking exceptions with friendly messages; added JavaScript to block past dates in the date picker. Aryan (Pharmacy): added validation for quantity, dose and vitals ranges; handled exceptions for stock below zero cases; added JavaScript validation to vitals form. Devkinandan (Billing): added validation for amount, payment mode and dates; handled exceptions for occupied room and duplicate payment; prevented overpayment greater than the bill total. All members shared their work on GitHub and discussed it in the team and in the Saturday mentor meeting.
Problems Faced	Devansh: Filter blocked static CSS and JS files. Deepak: Search became slow without proper query. Hansraj: Cancel was visible to wrong role. Aryan: Stock could go negative. Devkinandan: Overpayment case was missed. All problems were solved by team discussion and guidance from the mentor.
Plan for Next Week	Next week the team will move to: Module coding - module completion and self-review. Devansh: completed AdminServlet actions for user and department management. Deepak: completed the medical records view for the patient. Hansraj: completed consultation saving into medical records. Aryan: reduced stock automatically when a prescription is issued. Devkinandan: completed the admin reports.jsp with revenue and admission figures.

Week 9 | 31-08-2026 to 06-09-2026 | Module coding - module completion and self-review

Week dropdown	Week 9
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	This week the whole team the module was completed, reviewed and pushed to GitHub. Devansh (Authentication): completed AdminServlet actions for user and department management; added duplicate username and email validation in user creation; added activate and deactivate user option for admin. Deepak (Patient): completed the medical records view for the patient; added the doctor's records.jsp to view and add notes; added patient auto-ID generation. Hansraj (Appointment): completed consultation saving into medical records; added the status change to Completed after consultation; added filters on appointment lists by date and status. Aryan (Pharmacy): reduced stock automatically when a prescription is issued; added lab result status flow (Requested to Completed); added vital history list for each patient. Devkinandan (Billing): completed the admin reports.jsp with revenue and admission figures; added payment status filters to the bills list; added the notification entries on bill generation. All members shared their work on GitHub and discussed it in the team and in the Saturday mentor meeting.
Problems Faced	Devansh: Merge conflicts in shared files. Deepak: Time was short for comments and cleanup. Hansraj: Status did not update after consultation. Aryan: Lab status flow had gaps. Devkinandan: Report query was slow with many joins. All problems were solved by team discussion and guidance from the mentor.
Plan for Next Week	Next week the team will move to: Integration of all five modules. Devansh: merged all modules into one WAR project and resolved package conflicts. Deepak: integrated patient registration with the login of the patient user. Hansraj: integrated doctor list from Module 2 into booking. Aryan: linked prescriptions with consultation from Module 3. Devkinandan: linked bills to appointments and prescriptions from other modules.

Week 10 | 07-09-2026 to 13-09-2026 | Integration of all five modules

Week dropdown	Week 10
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	This week the whole team all five modules were integrated into one application. Devansh (Authentication): merged all modules into one WAR project and resolved package conflicts; connected the login redirect to every role dashboard; checked filter rules against every servlet URL pattern. Deepak (Patient): integrated patient registration with the login of the patient user; linked doctor listing with the appointment booking page; checked navigation between patient and doctor pages after merge. Hansraj (Appointment): integrated doctor list from Module 2 into booking; connected consultation with prescription and lab pages; fixed time format mismatch between JSP and DAO. Aryan (Pharmacy): linked prescriptions with consultation from Module 3; linked lab tests to the doctor's lab_reports page; linked patient prescriptions page with prescription data. Devkinandan (Billing): linked bills to appointments and prescriptions from other modules; linked admission with patient and room lists; connected the reports page to data from all modules. All members shared their work on GitHub and discussed it in the team and in the Saturday mentor meeting.
Problems Faced	Devansh: Sidebar links showed wrong items for some roles. Deepak: Connections were not closed in a few methods. Hansraj: Time format mismatch between JSP and DAO. Aryan: Foreign key errors after merge. Devkinandan: Totals mismatched after merging charges. All problems were solved by team discussion and guidance from the mentor.
Plan for Next Week	Next week the team will move to: Testing (unit, functional, role-based). Devansh: prepared test cases for login, wrong password and locked access. Deepak: wrote test cases for patient registration and profile update. Hansraj: wrote test cases for booking, rescheduling and cancelling. Aryan: wrote test cases for inventory and prescription. Devkinandan: wrote test cases for billing and payments.

Week 11 | 14-09-2026 to 20-09-2026 | Testing (unit, functional, role-based)

Week dropdown	Week 11
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	This week the whole team the application was tested in detail and defects were fixed. Devansh (Authentication): prepared test cases for login, wrong password and locked access; tested role-based access by opening restricted URLs directly; tested session expiry and logout behaviour. Deepak (Patient): wrote test cases for patient registration and profile update; tested invalid and boundary inputs on forms; tested doctor dashboard data with multiple doctors. Hansraj (Appointment): wrote test cases for booking, rescheduling and cancelling; tested double-booking prevention with same slot; tested consultation notes and record saving. Aryan (Pharmacy): wrote test cases for inventory and prescription; tested stock deduction with multiple medicines; tested lab request and result entry. Devkinandan (Billing): wrote test cases for billing and payments; tested tax and total calculations with samples; tested admission, room status and discharge. All members shared their work on GitHub and discussed it in the team and in the Saturday mentor meeting.
Problems Faced	Devansh: Session lost after server restart during tests. Deepak: Some inputs allowed negative age. Hansraj: A test case failed for same-time reschedule. Aryan: Edge case in stock deduction. Devkinandan: Rounding difference in a test case. All problems were solved by team discussion and guidance from the mentor.
Plan for Next Week	Next week the team will move to: Deployment (Docker, Render, cloud MySQL) and bug fixing. Devansh: wrote a Dockerfile (Maven build stage and Tomcat 10.1 run stage). Deepak: checked patient and doctor pages on the deployed site. Hansraj: tested booking flow on the live Render site. Aryan: checked inventory and prescription pages on the live site. Devkinandan: checked billing, admission and report pages on live site.

Week 12 | 21-09-2026 to 27-09-2026 | Deployment (Docker, Render, cloud MySQL) and bug fixing

Week dropdown	Week 12
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	This week the whole team the application was deployed to the cloud and bugs were fixed. Devansh (Authentication): wrote a Dockerfile (Maven build stage and Tomcat 10.1 run stage); created a cloud MySQL instance and moved the schema to it; updated db.properties for the cloud database with SSL. Deepak (Patient): checked patient and doctor pages on the deployed site; fixed date format issue on the cloud database; added missing seed data for departments and doctors. Hansraj (Appointment): tested booking flow on the live Render site; fixed time zone difference between local and cloud server; corrected status display bug on the appointments list. Aryan (Pharmacy): checked inventory and prescription pages on the live site; fixed foreign key insert order in seed data; corrected date display in the lab details page. Devkinandan (Billing): checked billing, admission and report pages on live site; fixed decimal formatting on the cloud database; corrected a missing room status default in seed data. All members shared their work on GitHub and discussed it in the team and in the Saturday mentor meeting.
Problems Faced	Devansh: Cloud DB SSL connection refused at first. Deepak: Date format differed on cloud MySQL. Hansraj: Time zone differed on cloud server. Aryan: Seed data failed on cloud DB. Devkinandan: Decimal formatting changed on cloud DB. All problems were solved by team discussion and guidance from the mentor.
Plan for Next Week	Next week the team will move to: Report, PPT, demo video and viva preparation. Devansh: wrote the introduction and tools sections of the project report. Deepak: wrote the patient and doctor module section of the report. Hansraj: wrote the appointment module section of the report. Aryan: wrote the pharmacy, lab and nursing section of the report. Devkinandan: wrote the billing, admission and reports section of the report.

Week 13 | 28-09-2026 to 04-10-2026 | Report, PPT, demo video and viva preparation

Week dropdown	Week 13
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	This week the whole team the report, presentation, demo video and viva preparation were completed. Devansh (Authentication): wrote the introduction and tools sections of the project report; documented authentication and security design in the report; prepared slides on architecture, roles and security. Deepak (Patient): wrote the patient and doctor module section of the report; added screenshots and table descriptions to the report; prepared module slides for the presentation. Hansraj (Appointment): wrote the appointment module section of the report; added flow diagrams and screenshots to the report; prepared slides on the appointment workflow. Aryan (Pharmacy): wrote the pharmacy, lab and nursing section of the report; added screenshots of inventory, prescription and vitals; prepared slides on pharmacy and lab workflow. Devkinandan (Billing): wrote the billing, admission and reports section of the report; added invoice and report screenshots; prepared slides on billing and admission workflow. All members shared their work on GitHub and discussed it in the team and in the Saturday mentor meeting.
Problems Faced	Devansh: Video recording took multiple attempts. Deepak: Report formatting took extra time. Hansraj: Exporting screenshots for report. Aryan: Many screenshots to collect. Devkinandan: Collecting output screenshots. All problems were solved by team discussion and guidance from the mentor.
Plan for Next Week	Final viva and demo of the working software. Submit the report, PPT, SRS, demo video and GitHub repository on the portal.